Karachi Institute of Economics and Technology
College of Computing and Information Sciences
FINAL YEAR PROJECT – Fall-2026
PROPOSAL

SEMESTER	FALL 2026	YEAR	2026-2027
Degree Program		BSCS/BSAI/BSSE/BSCYBSE	

TITLE OF PROPOSED PROJECT

Froggo : AI-Powered Frog Dissection Interactive Learning System

Project Category (choose one)	☐ Product based	☐ Research-based

SUPERVISOR INFORMATION (To be filled by the office)

Supervisor Name:		Organization/ Designation	
Contact No:		email:	

STUDENT(S) INFORMATION

S#	Student ID	Name	
1	64906	Mirza Abdul Samad	
Contact No:	03357547244	email:	samadmirza269@gmail.com
Expertise	Full Stack AI development		
2	15077	Anas Abid	
Contact No:	03172183886	email:	anasabid6009@gmail.com
Expertise	Web Development & Database Management		
3	65580	Ayman Nisa	
Contact No:	03141704686	email:	aymannisa.85@gmail.com
Expertise	Fronted and Backend Development		
4	65610	Muneeza Fayyaz	
Contact No:	03142250207	email:	muneezafayyaz252@gmail.com
Expertise	Fronted and Backend Development		

PROJECT AREA/TOOLS

Tools Required:

1.1.1 Programming Languages

- **Python** — AI, computer vision, image processing, backend logic
- **MySQL** — database queries if student progress/storage is included

1.1.2 AI / Machine Learning

- **PyTorch** — deep learning and action/gesture recognition
- **YOLO** — object/region detection where required
- **scikit-learn** — traditional ML algorithms and evaluation

	### 1.1.3 Computer Vision • **OpenCV** — camera processing, marker tracking, gesture extraction, perspective transformation, coordinate mapping ### 1.1.4 Frontend / User Interface • **PySide6 (Qt for Python)** — desktop application and digital learning interface ### 1.1.5 Backend • **FastAPI** — API/backend layer if you separate the AI engine from the application UI ### 1.1.6 Database • **SQLite** — lightweight local storage for students, sessions, actions and scores ### 1.1.7 Data Processing • **NumPy** • **Pandas** • **Pillow** ### 1.1.8 Development & Version Control • **Git** • **GitHub** • **VS Code** • **Jupyter Notebook**
Area/Specialization:	Artificial Intelligence & Computer Vision

SUMMARY OF PROPOSED PROJECT (MAXIMUM 400 WORDS)

Froggo is an interactive Frog Dissection learning system that uses computer vision to recognize actions performed by students on a transparent acrylic surface and reproduce them on a digital educational image in real time. Using a camera, OpenCV, and AI-based action recognition, the system detects gestures such as drawing, circling, selecting, highlighting, and marking, while accurately mapping physical actions to their corresponding digital locations. The system aims to provide a more

intuitive and engaging learning experience, initially focusing on biology and anatomy-based educational content.

PROJECT OBJECTIVE(S)/OUTCOMES

1. **To develop an AI-powered physical-to-digital interactive learning system** that enables students to interact with digital educational content through a physical transparent surface.
2. **To develop a computer vision system** capable of detecting and tracking marker-based interactions performed on the physical surface in real time.
3. **To implement AI-based action/gesture recognition** to identify user actions such as drawing, circling, selecting, highlighting, marking, and erasing.
4. **To accurately map physical actions to their corresponding digital coordinates**, ensuring that actions performed on the physical surface are reproduced at the correct location on the digital image.
5. **To provide real-time digital feedback** by reflecting the user's physical actions on educational content displayed on a computer screen.
6. **To evaluate the system's performance** based on action-recognition accuracy, coordinate-mapping accuracy, and response time.

SIMILAR APPLICATIONS (SIMILARITY/DISSIMILARITY)

S#	Name of App	Link (URL/Play Store)	Similar Features	New Features
1	**Osmo**	N/A	Uses a camera to recognize physical objects/actions and connects **physical interaction with digital content** on a screen.	**Froggo** focuses on **AI-based recognition of marker gestures/actions on a transparent acrylic surface** and maps those actions directly onto educational images. It is designed specifically for educational diagrams and can support actions such as drawing, circling, selecting, highlighting and marking.

2	Virtual Frog Dissection	N/A	Provides a digital frog-dissection learning experience with virtual dissection tools, organ information, labels and step-by-step learning	Unlike a conventional virtual dissection application, **Froggo** allows the **student to physically perform actions on a transparent surface**. Computer vision recognizes the physical action and reproduces it digitally. **Froggo** can therefore act as a bridge between **hands-on interaction and digital learning**.
3	**Merge EDU / Merge Explorer**	Merge EDU	Combines hands-on physical interaction with digital educational experiences and provides interactive science learning, including biology and frog-related content	Merge primarily uses **AR/3D experiences and Merge hardware**, whereas **Froggo** uses a **simple transparent acrylic board + camera**. **Froggo** focuses on **2D educational content, AI gesture recognition, physical-to-digital coordinate mapping, and real-time reproduction of user actions** without requiring AR/VR or 3D models.

Sustainable Development Goals (SDG)

1. ☐ No Poverty
2. ☐ Zero Hunger
3. ☐ Good Health and Well-being
4. ☐ Quality Education
5. ☐ Gender Equality
6. ☐ Clean Water and Sanitation
7. ☐ Affordable and Clean Energy
8. ☐ Decent Work and Economic Growth
9. ☐ Industry, Innovation, and Infrastructure
10. ☐ Reduced Inequalities
11. ☐ Sustainable Cities and Communities
12. ☐ Responsible Consumption and Production
13. ☐ Climate Action
14. ☐ Life Below Water
15. ☐ Life on Land
16. ☐ Peace, Justice, and Strong Institutions
17. ☐ Partnerships for the Goals

FUNCTIONAL FEATURES (Must be in Bullet Form)

For FYP 1 (At Least 70% of FYP):

Froggo contributes primarily to SDG 4: Quality Education by developing an interactive and technology-driven Frog Dissection learning system that makes complex biological concepts more engaging, visual, and intuitive. The project combines Artificial Intelligence, Computer Vision, and Human-Computer Interaction to allow students to interact with physical educational material and receive corresponding digital visualizations and feedback. This approach can improve understanding of subjects such as biology and anatomy while providing a more interactive alternative to conventional learning methods.

The project also supports SDG 9: Industry, Innovation and Infrastructure through the development of an innovative AI- and computer-vision-based educational interface.

For FYP 2:

FYP 2 will further enhance Froggo by integrating and refining the AI-based recognition, computer vision, physical-to-digital interaction, and real-time visualization components into a functional educational prototype. The completed system will demonstrate how AI and emerging interactive technologies can be applied to improve practical and experiential learning.

The project will continue contributing to SDG 4 (Quality Education) and SDG 9 (Industry, Innovation and Infrastructure), with SDG 10 (Reduced Inequalities) being considered if the final system emphasizes affordability and accessibility for institutions with limited educational resources.

Additional Features by Jury Members:

For Proposal Defense Purpose

PROPOSED ADVISORY COMMITTEE		
S#	**Faculty Name**	**Signature**
1		
2		
3		
4		

FYP COMMITTEE			
S#	**Member(s) Name**	**Designation**	**Signature**
1	**Mr. Usman Khan**	Head, Final Year Projects	
2	**Dr. Maaz Bin Ahmed/ Dr. Sohail Imran/ Dr. Muhammad Fahad/ Dr. Ayub Latif**	Head of Dept.	
3	**Prof Dr Muhammad Khalid Kham**	Dean CIS	
	Date	00/00/0000	